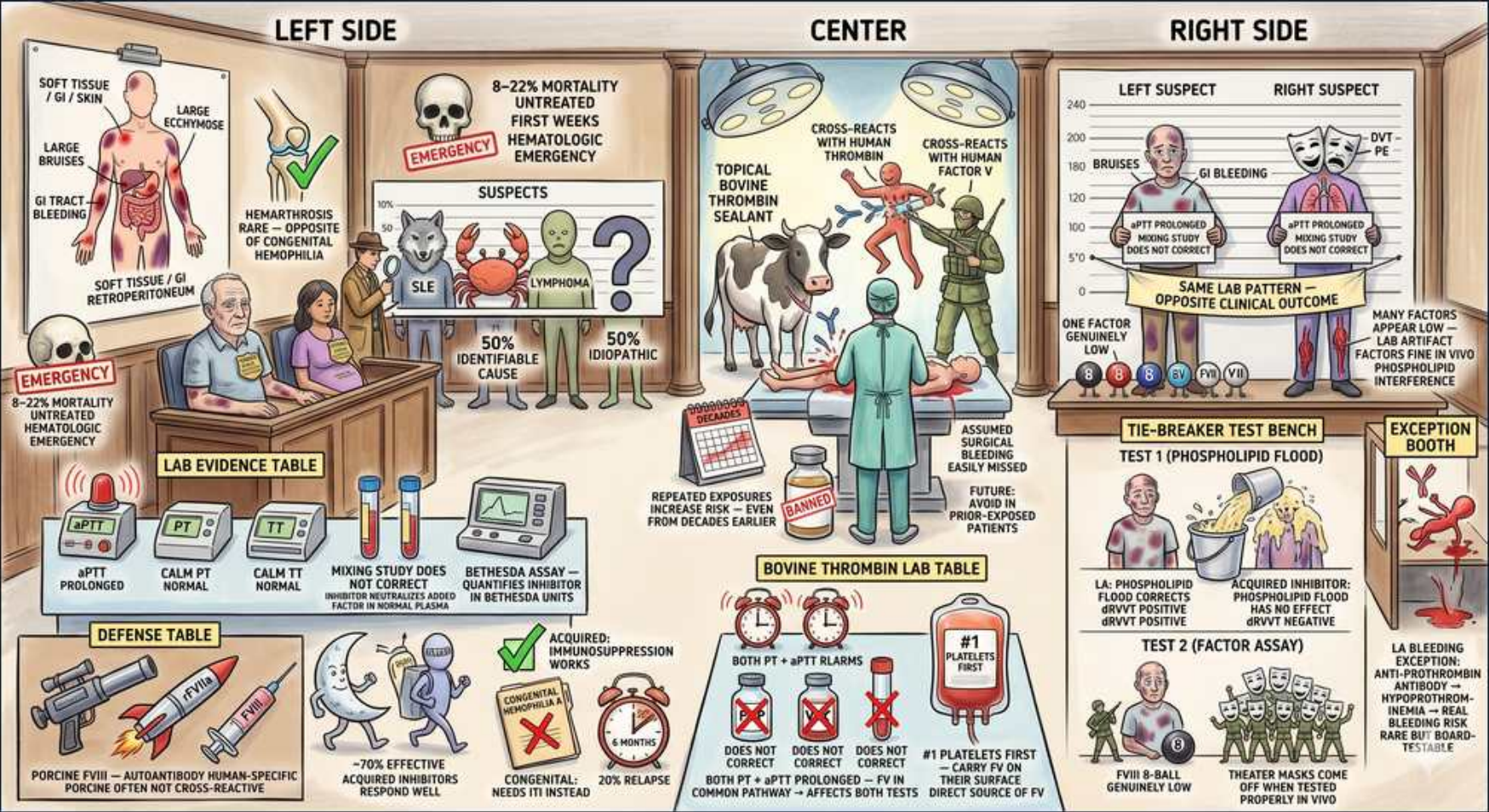


Coagulation — Acquired Inhibitors of Coagulation Factors



1. Acquired hemophilia A

WHAT YOU SEE

Left side body 'large bruises, soft tissue / GI / GU bleeding'

WHAT IT ENCODES

Acquired hemophilia A = a spontaneously arising IgG AUTOantibody (most often IgG4) directed against factor VIII in a person with no prior bleeding history. The antibody neutralizes FVIII activity, producing new-onset, often dramatic bleeding. Classic board vignette: an elderly patient (or a postpartum woman) with sudden extensive ecchymoses and a long isolated aPTT. Mnemonic: 'old person + new bruising + long aPTT that won't mix = acquired hemophilia A.'

BOARD TRAP

It is an ACQUIRED AUTOantibody, NOT an inherited factor deficiency and NOT an alloantibody from prior factor infusions (alloantibodies arise in congenital hemophiliacs treated with FVIII). Sex doesn't matter — unlike X-linked congenital hemophilia A, acquired disease affects men and women equally.

2. Soft tissue / mucosal bleeding

WHAT YOU SEE

Left 'soft tissue/GI/GU/retroperitoneal bleeding' icons

WHAT IT ENCODES

The bleeding phenotype is soft-tissue, mucosal, GU/GI, intramuscular, and retroperitoneal bleeding plus large spreading ecchymoses — the OPPOSITE distribution of inherited hemophilia. This pattern reflects an acquired soft-tissue/mucocutaneous bleeding tendency rather than the deep joint bleeding of congenital deficiency.

BOARD TRAP

Hemarthrosis (joint bleeding), the hallmark of CONGENITAL hemophilia A, is characteristically RARE in acquired hemophilia A. On the exam, 'spontaneous hemarthrosis since childhood' = congenital; 'new soft-tissue/mucosal bleeds in an adult' = acquired.

3. Hematologic emergency

WHAT YOU SEE

Left skull 'emergency, 8-22% mortality untreated, first weeks'

WHAT IT ENCODES

Acquired hemophilia A is a true hematologic emergency with ~8–22% mortality, and most fatal bleeds occur in the FIRST weeks after presentation. Early recognition and prompt hemostatic control change outcomes. Mnemonic: the skull = 'don't wait — bleed-out risk is front-loaded.'

BOARD TRAP

Do NOT undertreat or dismiss it as benign bruising — it can be rapidly fatal. The danger is highest early, before the inhibitor has been eradicated, so hemostatic therapy starts immediately rather than waiting for immunosuppression to work.

4. Associations (malignancy, autoimmune, postpartum)

WHAT YOU SEE

Center-left suspects 'SLE, lymphoma, malignancy, postpartum, idiopathic ~50%'

WHAT IT ENCODES

Triggers (the 'usual suspects'): autoimmune disease (SLE, RA), malignancy (solid tumors and lymphoproliferative disorders), the postpartum state, and certain drugs. Roughly 50% are IDIOPATHIC, and these cluster in the elderly. Mnemonic: 'CAMP — Cancer, Autoimmune, Medications, Postpartum/idioPathic.'

BOARD TRAP

No identifiable trigger is COMMON — being idiopathic does not exclude the diagnosis. Conversely, a new acquired-hemophilia diagnosis should prompt an age-appropriate malignancy/autoimmune work-up rather than reassurance.

5. aPTT prolonged, PT normal

WHAT YOU SEE

Lab evidence table bottom-left 'aPTT prolonged, PT normal, TT normal'

WHAT IT ENCODES

The screening pattern is an ISOLATED prolonged aPTT with NORMAL PT/INR, normal thrombin time, and normal platelet count — because anti-FVIII hits only the intrinsic pathway. This looks identical to hemophilia A on screening labs until you do a mixing study. Mnemonic: 'lonely long aPTT.'

BOARD TRAP

A normal PT with an isolated long aPTT mimics a simple factor deficiency (FVIII, IX, XI, XII) — you cannot tell deficiency from inhibitor on the screen alone; the mixing study is the next step.

6. Mixing study does NOT correct

WHAT YOU SEE

Center-bottom 'mixing study does not correct, monitor incubate 1-2h'

WHAT IT ENCODES

KEY DISTINGUISHER: mix patient plasma 1:1 with normal plasma. In factor DEFICIENCY the aPTT CORRECTS (normal plasma supplies the missing factor). With an INHIBITOR it fails to correct, and after incubation 1–2 h at 37°C the aPTT actually WORSENS — a time- and temperature-dependent inhibitor classic for anti-FVIII. Mnemonic: 'deficiency fills the gap; inhibitor eats the donor too — and gets worse with time.'

BOARD TRAP

If mixing CORRECTS → factor deficiency; if it FAILS to correct → inhibitor. The progressive worsening on incubation specifically fingers a FVIII autoantibody (a slow, time-dependent inhibitor), distinct from the immediate inhibitor pattern of a lupus anticoagulant.

7. Bethesda assay quantifies

WHAT YOU SEE

'Bethesda assay quantifies inhibitor in Bethesda units'

WHAT IT ENCODES

The inhibitor strength is measured by the Bethesda assay; 1 Bethesda Unit (BU) = the amount of inhibitor that neutralizes 50% of FVIII in 2 h at 37°C. Titer guides therapy: low titer (<5 BU) may respond to high-dose FVIII; high titer (>5 BU) requires bypassing agents.

BOARD TRAP

You must measure the specific FVIII activity AND the Bethesda titer — the aPTT alone neither confirms the diagnosis nor guides treatment. High Bethesda titer is why simply giving more FVIII fails.

8. Factor VIII (most common target)

WHAT YOU SEE

A center panel 'ONE factor low' pointing at the FVIII suspect — a single isolated low factor with a non-correcting mix = a specific inhibitor

WHAT IT ENCODES

Acquired inhibitors most often target factor VIII (hence 'acquired hemophilia A'); the result is a SINGLE low factor activity with a non-correcting mixing study. Inhibitors to other factors (V, IX, XI) exist but are far rarer. Mnemonic: 'one factor down, mix won't fix = inhibitor.'

BOARD TRAP

MANY factors low at once suggests a GLOBAL problem — liver disease, DIC, or vitamin K deficiency — whereas ONE isolated low factor with a non-correcting mix points to a specific inhibitor. Don't reflexively call a multi-factor pattern an 'inhibitor.'

9. Lupus anticoagulant mimic

WHAT YOU SEE

A right-side suspect 'aPTT prolonged, mixing won't correct — phospholipid-DEPENDENT interference' — the lupus-anticoagulant mimic that clots instead of bleeds

WHAT IT ENCODES

The great mimic: a lupus anticoagulant (LA) also prolongs the aPTT and also fails to correct on mixing, but it is phospholipid-DEPENDENT (an antiphospholipid antibody) and clinically causes THROMBOSIS and pregnancy loss — not bleeding. Same lab screen, opposite clinical picture. Mnemonic: 'LA = Lupus Anticoagulant = Lots of clots, not a leak.'

BOARD TRAP

Lupus anticoagulant → CLOTTING (arterial/venous thrombosis, recurrent fetal loss), NOT bleeding. If an LA patient bleeds, look for a separate cause (e.g., associated thrombocytopenia or hypoprothrombinemia–LA syndrome). Never anticoagulate by mistake assuming a 'bleeding' inhibitor.

10. Phospholipid flood test (LA)

WHAT YOU SEE

A right tie-breaker bench 'TEST 1 — phospholipid flood corrects LA' — excess phospholipid mops up a lupus anticoagulant but not a true factor inhibitor

WHAT IT ENCODES

Tie-breaker test #1: flood the assay with EXCESS phospholipid (the phospholipid-neutralization/correction step, e.g., the DRVVT confirm or hexagonal-phase assay). Adding phospholipid OVERWHELMS and CORRECTS a lupus anticoagulant but does NOT correct a true factor inhibitor — because the autoantibody attacks the factor, not the phospholipid. Mnemonic: 'phospholipid mops up the LA.'

BOARD TRAP

Correction with added phospholipid = lupus anticoagulant. Failure to correct with added phospholipid (while still failing the 1:1 mix) = a genuine factor inhibitor such as anti-FVIII.

11. Factor assay tie-breaker

WHAT YOU SEE

A right tie-breaker 'TEST 2 — factor assay, FVIII genuinely low' with an 8-ball — a real inhibitor gives a dilution-stable truly low FVIII

WHAT IT ENCODES

Tie-breaker test #2: the specific factor assay. In acquired hemophilia A the FVIII activity is GENUINELY and reproducibly low. With a lupus anticoagulant, several factor assays appear FALSELY low at standard dilution but normalize (rise toward normal) when the patient plasma is further DILUTED — a key feature exposing an artifact rather than a true deficiency.

BOARD TRAP

A lupus anticoagulant can ARTIFACTUALLY lower multiple factor assays simultaneously; performing the assay at increasing dilutions unmasks the artifact. A single, dilution-stable, truly low FVIII points to a real inhibitor, not LA interference.

12. Bypassing agents for bleeding

WHAT YOU SEE

The bovine-thrombin/bypass area 'rFVIIa or aPCC — #1 platelets first' — agents that generate thrombin around the blocked FVIII

WHAT IT ENCODES

Acute bleeding control in high-titer acquired hemophilia A: use BYPASSING agents that work downstream of FVIII — recombinant activated factor VII (rFVIIa, NovoSeven) or activated prothrombin complex concentrate (aPCC / FEIBA). These generate thrombin without needing FVIII, so the inhibitor titer doesn't block them. (Emicizumab, a FVIII-mimetic bispecific antibody, is an emerging option.) Mnemonic: 'go AROUND the blocked factor.'

BOARD TRAP

Giving plain FVIII concentrate FAILS against a high-titer inhibitor (it is rapidly neutralized). Only low-titer disease may respond to high-dose FVIII or DDAVP; high-titer disease requires bypassing agents.

13. Immunosuppression eradicates

WHAT YOU SEE

The bottom-left defense table 'acquired -> immunosuppression works, ~70% effective' — steroids/cyclophosphamide/rituximab eradicate the autoantibody

WHAT IT ENCODES

Definitive long-term therapy is ERADICATION of the autoantibody with immunosuppression — first-line corticosteroids ± cyclophosphamide, with rituximab as an alternative/add-on; remission is achieved in roughly 70%, though relapse (~20%) occurs. This is the feature that separates ACQUIRED from congenital hemophilia — you can remove the cause. Mnemonic: 'bypass to stop the bleed NOW, immunosuppress to cure it.'

BOARD TRAP

Bypassing agents only treat the active bleed — they do NOT remove the antibody. Immunosuppression is required to eradicate the inhibitor. (Congenital hemophilia, by contrast, is not immunosuppressed — there is no autoantibody to eradicate; it is treated with factor replacement/prophylaxis.)